Wenjie Luo

Operations Planning | Industrial Engineering | Decision Optimization | Graduating Jan 2027

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PROFILE

Turn operational questions into metrics, plans and optimization models. Apply SQL, LP/MIP, forecasting and process analysis, with experience in supplier, materials and shipment coordination; seeking planning and operations-improvement roles in semiconductor manufacturing.

EDUCATION

National University of Singapore | MSc Civil Engineering (Transportation) Aug 2025 - Jan 2027
Transport focus; research in transportation information and control.

Chang'an University | BEng Transportation Engineering Sep 2021 - Jun 2025
GPA 3.16/5.0; approximately top 15% of the cohort.

EXPERIENCE

Far East Facade | Project Engineer, Engineering Department Jun 2026 - Present
- Coordinate testing and deliveries for 15 façade material types across 5 suppliers; track milestones on a status board and parse supplier emails in Python.

Chery International | Product Intern May 2026 - Jun 2026
- Produced sales forecasts and a Gantt plan for KD (knock-down) production and logistics, aligning milestones with shipments of sample vehicles and production parts.
- Created 15 personas and 5 VOC reports; used competitor, price and registration data to inform product positioning.

Expeditors | Supply Chain Assistant Dec 2024 - Apr 2025
- Improved document-processing spreadsheets with VLOOKUP and pivot tables, raising efficiency by approximately 30%; refined existing standardized checks to reduce manual errors.

PROJECTS

LP/MIP Modeling and Solver Experiments Sep 2026
- Built an LP/MIP modeling layer and solver; checked 11 cases against HiGHS. Assignment-style models solved 6–12 trip instances at the LP relaxation (1 search node each); big-M models still needed 21–31 nodes on smaller 4–6 trip instances.
- Validated pure-integer Gomory cuts on 5,376 binary points. Three knapsack cases needed only 1 search node; a counterexample rose from 29 to 31 nodes, showing that cuts can add work.

Malaysia Automotive Metrics and Data Assistant Aug 2026
- Defined wholesale, registration and market-share metrics with SQL templates; linked demo-warehouse outputs to source tables through a 14-node, 20-edge lineage graph.
- Answered 22 of 30 authored questions correctly and requested clarification on 8; tested clarification checks by ablation.

Lean Construction | Course Project Feb - Mar 2026
- Contributed to a video study of three formwork cycles; mapped process flow and crew balance, identifying 297 seconds of waiting (about 46%).
- Proposed smaller batches and a revised layout without an intermediate buffer; calculated a cycle time of 6 min 40 sec versus about 10 minutes.

SKILLS

Analytics: Python, SQL, Excel (VLOOKUP, pivot tables), metric definitions, data lineage, forecasting
Methods: LP/MIP, branch-and-bound, process mapping, time-and-motion studies, Gantt planning, paired bootstrap